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Hierarchically porous hydrogel electrolyte prepared from interpenetrating polymer networks for flexible Zn-Air batteries

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ABSTRACT

Gel polymer electrolyte (GPE) based flexible zinc-air batteries (ZABs) are considered promising power sources for next-generation wearable devices because of their high specific energy density, low cost, high safety, and environmental friendliness. However, the liquid component in GPE is susceptible to evaporation through the air cathode, drastically reducing the cell performance and lifetime. Poly(vinyl alcohol) (PVA) is the most widely adopted ion conductive polymer, but its poor water retention and low ionic conductivity hinder the widespread commercialization of flexible ZABs. A hierarchically porous (HP) PVA and poly(acrylic acid) (PAA) hybrid GPE is developed to address the aforementioned issues. The polymerization of acrylic acid in a PVA matrix forms a semi-interpenetrating polymer network involving a unique HP structure inside the membrane. The PAA-rich lotus-seed-like pores fundamentally improve electrolyte uptake and retention properties, and enhance ionic conductivity. The HP-PVA/PAA hybrid GPE delivers excellent cycling stability, rate characteristics, and power output in ZABs. Furthermore, HP-PVA/PAA based flexible ZABs incorporated into the customized strap demonstrates their feasibility for powering the wearable smart watch.

1. Introduction

Due to the ever-developing integrated internet of things and artificial intelligence technologies, wearable technology continues to develop new electronic devices designed to be worn on the human body that aim to assist with daily activities, monitor the current health status of the user, and provide prompt turn-by-turn directions [1–3]. Moreover, considerable efforts have been devoted over the past few years to developing a customized energy supply with sufficient energy density, flexibility, and user-safety. As next-generation wearable power sources, Zn-air batteries (ZABs) are considered the most promising candidates for wearable devices owing to their high theoretical energy density (1086 Wh kg^{-1}), low cost, high safety, and environmental friendliness [4,5]. Flexible ZABs usually consist of Zn metal as the anode, a porous carbonaceous air cathode, and a semi-solid state gel polymer electrolyte (GPE) as the main compartments, which are assembled with flexible current collectors and packaging materials [6–8].

Usually, GPEs contain a hydrogel polymer matrix soaked with alkaline electrolytes owing to their practically acceptable ionic conductivity in the range of 10^{-4} to 10^{-3} S cm^{-1} at ambient temperature [9]. Poly(vinyl alcohol) (PVA), a hydrophilic polymer, is commonly employed as a host polymer because of its excellent chemical and mechanical stability, nontoxicity, and easy processability [10]. However, the unavoidable evaporation of liquid components through the "open cathode" to the ambient air significantly reduces the ionic conductivity of PVA based GPEs during battery operation and severely limits the cycle life of the prototype cells [11–15]. Considerable attempts have been made by some authors to seek other polymer based GPE systems or

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organic/inorganic additives that improve the limitations of PVA based GPEs. For example, poly(acrylamide) has been reported as a promising candidate owing to its ionic conductivity and mechanical stability [16]. Similarly, cellulose-based materials, such as bacterial cellulose, have also been studied as potential solid-state electrolytes for ZABs [17,18]. These alternative electrolyte materials offer advantages over hydrogel polymer matrix-based electrolytes, including improved stability, reduced evaporation of liquid components, and increased ionic conductivity at room temperature. However, further research is required to fully understand the performance of these alternative electrolyte materials in ZABs, including their limited alkaline-tolerance, unsatisfactory water retention capability, low solubility, and the resistance to being reshaped.

Poly(acrylic acid) (PAA), a superabsorbent polymer, is a representative anionic polymer electrolyte that has attracted attention owing to its high alkaline-tolerance, excellent water retention, and ionic conductivity [19,20]. Tran et al. reported that the concentration of the precursor, crosslinkers, and monomers significantly affects the polymerization to form PAA based GPEs and the performance of ZABs [21]. Recently, Song *et al.* fabricated a hybrid semi-solid GPE containing two polymers (PVA and PAA), graphene oxide (GO), and potassium iodide as a reaction modifier to enhance the properties of the GPE. They suggested that the formation of a highly porous, wrinkled microstructure and the hydrogen bonding between the polymer chains and oxygen functional groups of GO improved the water retention capability of the hydrogel. Porous PVA based GPEs were also prepared to increase the pore volume for holding water through the selective dissolution mechanism of soluble poly(ethylene glycol) and silica additives, exhibiting a high conductivity (57.3 mS cm^{-1}), which improved the water retention capacity [22].

It has been widely accepted that porous-structured PVA and hydroxyl group-functionalized fillers are expected to improve the overall water absorption and electrolyte retention capability of GPE, but the introduction of sacrificial polymers or inorganic additives increases the complexity of the manufacturing process. Furthermore, the unsatisfactory open pore structure on the surface layer of the PVA matrix can facilitate the re-evaporation of the electrolyte, so a tailored design of a porous structure inside the membrane is highly desirable. To the best of our knowledge, the establishment of porous GPEs based on thermodynamic polymer chemistry without the introduction of other additives has never been reported for ZAB applications. The objective of this study was to design a porous GPE with superior ionic conductivity and electrolyte retention capability using a simple but practical fabrication process.

An interpenetrating polymer network (IPN) is a unique combination of two or more interwining crosslinked polymers, in which one of the network polymers is crosslinked in the presence of the other. This is distinct from other polymer combinations such as simple blends or copolymers. IPNs exhibit two distinct characteristics: first, the combined IPN has the advantages of both polymers; second, the IPN morphology can be adjusted under specific polymerization conditions. The conditions that may be adjusted during the polymerization process include the temperature, pressure, catalyst concentration, solvent, and crosslinking density; the adjustments enable the formation of a dispersion phase up to the micrometer scale [23–26]. Although IPNs have been extensively studied as candidates for various hydrogels and solid-state electrolytes, [27] their potential for use as porous GPEs in ZAB applications has rarely been investigated. Moreover, the properties of IPNs with regard to their polymerization conditions, including their morphology, porosity, mechanical strength, and electrochemical properties, have not been thoroughly explored.

In this study, we propose a hierarchically porous PVA and PAA (HP-PVA/PAA) membrane that outperforms conventional PVA based GPE systems. The semi-IPN formation method was employed, in which crosslinked PAA chains were generated in an aqueous solution by crosslinking the AA monomer within the PVA matrix; this resulted in a distinctive closed-pore structure resembling a lotus seed. The semi-IPN was implemented by phase separation during the polymerization of AA in a PVA matrix without any additives, resulting in lotus-seed-like closed pore structure. These unique pores can act as numerous water resources for a long period of time and facile ionic conductive channels. The structural formation mechanism was investigated using molecular dynamics (MD) simulations and polymer thermodynamics combined with microstructural observations. After the infiltration of a 6 M KOH solution, the obtained HP-PVA/PAA GPE delivered superior ionic conductivity (212 mS cm-1), which was attributed to the enhanced chain segmental motion in the semi-IPN and the numerous ion diffusion channels through the absorbed liquid electrolyte in the internal pores of membrane. Owing to these advantages, the assembled ZAB exhibited excellent cycling stability, discharge performance, and rate characteristics. We further demonstrate that sandwich-type flexible ZABs integrated into a bendable strap successfully serve as an additional power supply for a conventional smart watch.

2. Results and discussion

The overall fabrication procedures for the HP-PVA/PAA-based GPE and proposed ZAB cell are shown in Fig. 1. Specifically, AA monomers, crosslinkers (triallylamine), and neutralizers (KOH) were first introduced into the pre-dissolved PVA solution. After stirring for several hours, the AA monomers and crosslinkers were homogeneously distributed among the linear PVA chains. The polymerization of the AA monomers was then initiated by the addition of an initiator (potassium persulfate, KPS), which led to phase separation between the interpenetratively-polymerized PAA and PVA matrix, the so-called semi-IPN formation [24,25,28-30]. The polymerization process turned the mixed solution milky and opaque. Then, the mixed solution was poured into a Petri dish and dried to obtain a film-type HP-PVA/PAA membrane. The membrane thickness was tuned by varying the amount of poured solution. The obtained milky-colored membrane exhibited high flexibility and could be easily rolled up, as shown in the digital photographs. Interestingly, the resulting membrane exhibited a unique spherically porous structure, similar to lotus seeds (see cross-view scanning electron microscopy (SEM) and inset photographs in Fig. 1). It should be noted that the surface of the membrane was still smooth and dense, while pores were only observed in the cross-sectional view of the membrane (Fig. S1). This means that hierarchical pores were formed inside the membrane during polymerization. The obtained HP membrane was then soaked in a 6 M KOH solution for use as a GPE in flexible ZABs [19,22,31]. The impregnated GPE was sandwiched between a carbon cloth (CC) as an air cathode and a Zn foil anode. The HP-PVA/PAA GPE provided high liquid electrolyte uptake for a long period of time and high ionic conductivity during cell operation because of the numerous pore channels. Owing to the aforementioned advantages, the HP-PVA/PAA GPE used ZAB is expected to show excellent cell performance and long-term stability, which are significantly better than those of pristine PVA based GPEs or conventionally blended PVA+PAA GPEs.

The microstructural evolution of the developed HP-PVA/PAA is due to the thermodynamic immiscibility that appears during the fabrication of semi-IPN and leads to phase separation of the two polymers [25,26, 30]. To verify the underlying mechanism of the phase separation between the PAA and PVA molecules, MD simulations were performed, as shown in Figs. 2a and S2. PVA, water molecules, potassium ions (K^+, derived from the KOH neutralizer), and four different lengths of PAA molecules (from PVA/PAA-1 to PVA/PAA-4) were considered to describe the polymerization progress. Computational details, including the equilibration process and interaction energy calculation method, are described in Supporting Information and Table S1. The molecular distribution of K^+/PAA and PVA was compared on three axes (x, y, and z-directions) (Fig. S3). After MD simulations for 20 ns at 100 °C, the separation behavior of K^+/PAA and PVA was more clearly observed in

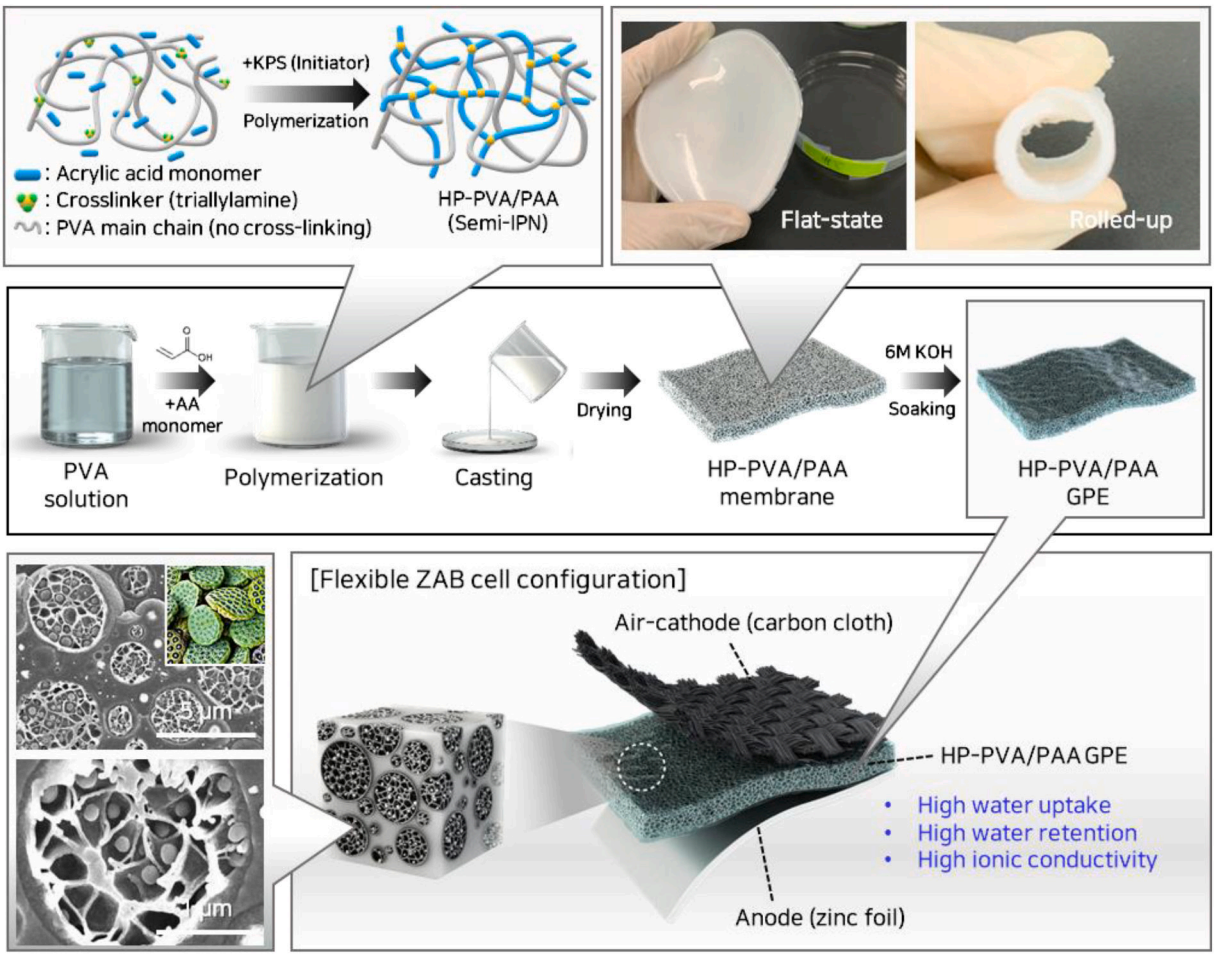


Fig. 1. Schematic of the fabrication process of the HP-PVA/PAA based GPE and flexible ZAB cell configuration.

the PVA/PAA-4 model than in the PVA/PAA-1 model. Fig. 2b summarizes the interaction energies between the K^+ ion and PAA (green bars) and K^+/PAA and PVA (gray bars) during the polymerization of PAA. The interaction energy quantifies the non-bond interaction between the constituent atoms, and the smaller the value, the stronger the interaction. As PAA polymerization proceeds, the AA with carboxylate groups (COO-) is covalently bonded and negative charges are concentrated near the chain, forming a strong electrostatic attraction with K^+. Subsequently, the K^+ adsorbed on PAA attracts other neighboring acid groups and induces the PAA chains to self-aggregate. On the other hand, the PVA chain is excluded from the electrostatic interactions posed by K^+, and thus it becomes difficult to access the K^+/PAA cluster as the polymerization of AA proceeds, as described in Fig. S4. The simulation results clearly provide thermodynamic evidence of polymeric phase separation between PVA and PAA during the semi-IPN formation process.

The phase diagram (including the three compositions, PVA, AA, and PAA in the triangle map, and temperature in the z-axis) was further studied, and the observed microstructural results from the different processing conditions were compared. The phase separation behavior during polymerization of the AA monomer in the PVA matrix was determined by the molecular fraction components, temperature, and polymerization time. The phase separation of semi-IPN essentially follows two mechanisms, namely, i) nucleation and growth mechanisms that occur in the metastable region (between the spinodal and binodal curves) and ii) spinodal decomposition, which occurs in the unstable region [25,26,32-34]. As shown in Fig. 2c, a modified phase diagram was compiled based on previous literature, and the microstructural formations were proposed using our SEM observation results under various conditions [31–34]. The initial ratio of PVA to AA monomers determines the phase separation pathways, that is, nucleation and growth (red line) and spinodal decomposition (orange line), to form semi-interpenetrating PVA and PAA networks.

The nucleation and growth mechanisms result in the phases being separated by the growth of nuclei, with one polymer-poor phase disseminated into the polymer-rich matrix as spherical droplets. As the process proceeds, the spherical droplets grow into a dispersed phase with a closed pore structure. In stage I, the AA monomers and cross-linkers (with a PVA and AA weight ratio of 2:1) are present in the linear PVA polymer chains. Phase separation starts in the metastable region (between the spinodal and binodal curves) of the phase diagram (stage II) [33]. Metastable and binodal demixing proceeds at the interfaces between the polymer (PVA)-lean and polymer-rich phases, followed by the nucleation and growth mechanism of a dispersed minor component, typically resulting in the formation of droplets (stage III) [26]. More specifically, pure PAA nodules are formed and propagate into the PVA matrix with an increasing AA oligomer molecular weight, as illustrated in stages II and III. This leads to phase separation, in which spherical domains of one of the polymers (PAA) are dispersed in a matrix of the other polymer (PVA) [29]. Finally, the phase separation is terminated with a certain polymer molecular weight, producing the bicontinuous spherical domains of the PAA-rich phase in the PVA matrix (stage IV). Fig. S5 shows sequential SEM images of the HP-PVA/PAA membrane at the various stages from I to IV, and further images (different magnifications in the cross-sectional view).

In addition, when the compositional ratio of PVA and AA was changed to 1:1, the phase separation occurred in the unstable region, resulting in spinodal decomposition in stages V and VI. The time

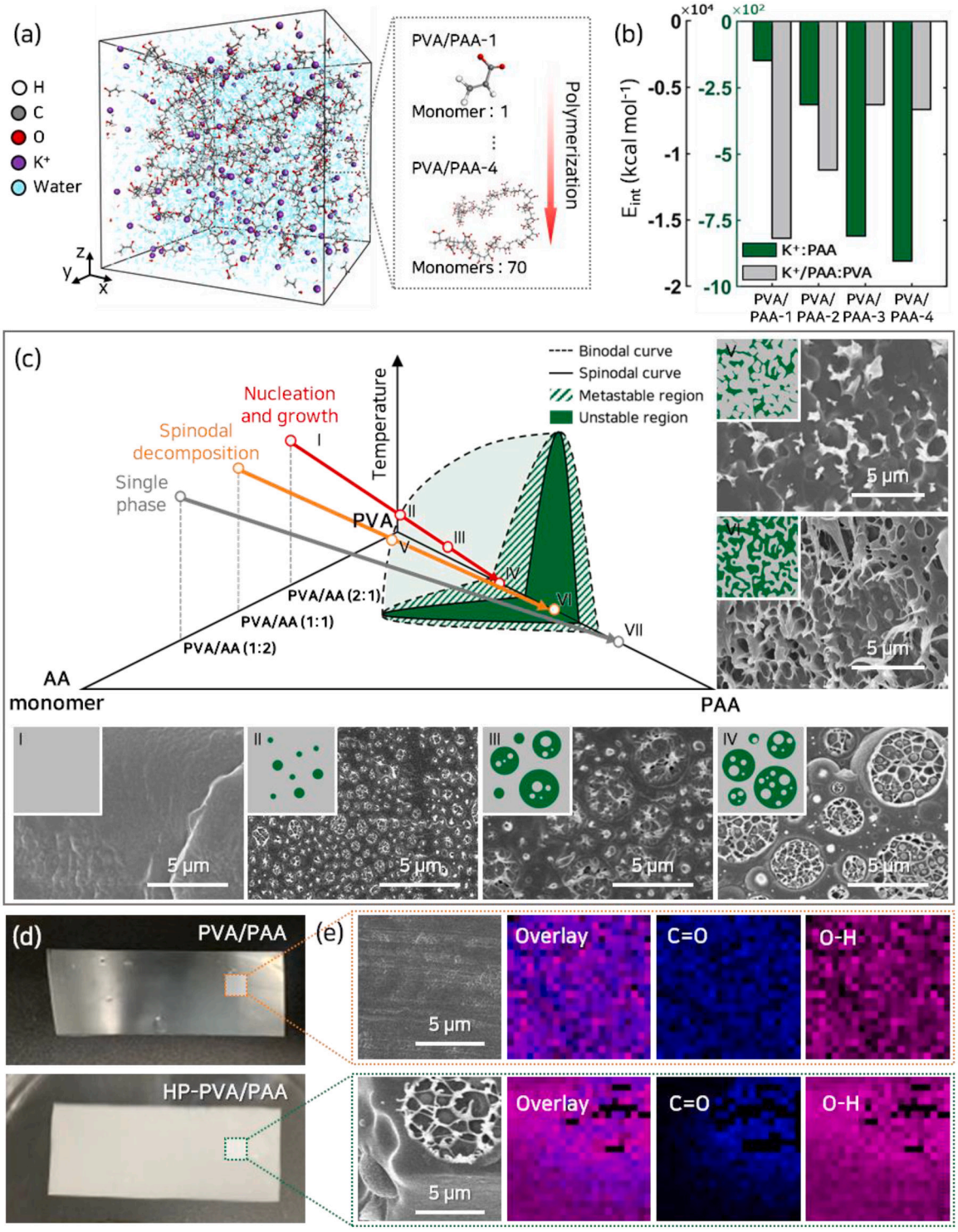


Fig. 2. (a) PVA/PAA unit cells modeled using all-atom MD simulations and (b) computed interaction energies between K^+ and PAA and K^+/PAA and PVA. (c) Proposed PVA/AA/PAA-based IPN phase transition diagram, and SEM images of the PVA/PAA based semi-IPN under different conditions. (d) Optical images and (e) Raman mapping of the blended PVA+PAA and HP-PVA/PAA membranes.

evolution of the concentration fluctuations, which normally triggers phase separation, resulted in a spatial wavelength and a bicontinuous 3D structure until phase hardening [32–34]. Fig. S6 further shows the microstructural evolution of the spinodally-decomposed PVA/PAA membrane. In contrast, when the composition ratio of PVA to AA was changed to 1:2, the semi-IPN polymerized as a single-phase membrane in the stable region without phase separation, resulting in stage VII (Fig. S7).

To verify the mechanical strength and ensure their suitability for use as a GPE, we conducted tensile strength tests on membranes containing pure PVA, physically blended PVA+PAA (2:1), and semi-IPN PVA/PAA membranes with ratios of 2:1, 1:1, and 1:2. As shown in Fig. S8, the yield points of the blended PVA+PAA and semi-IPN PVA/PAA membranes were lower than those of pure PVA, indicating a deterioration in the mechanical properties as the percentage of PAA increased in the membrane [35–37]. Moreover, the semi-IPN PVA/PAA (1:2) was easily

broken when placed in the instrument and its tensile strength could not be measured. Nevertheless, even when the PVA:PAA ratio was the same (2:1), the semi-IPN PVA/PAA displayed a higher mechanical strength than that of the physically blended PVA+PAA owing to the formation of an interpenetrating network structure. Thus, considering the GPE requirements, the HP-PVA/PAA membrane (PVA:AA = 2:1) with closed pore from the nucleation and growth pathways was adopted for further characterization.

For comparison, pristine PVA and physically blended PVA+PAA membranes were prepared, and their SEM images are presented in Figs. S9 and S10, respectively. Fig. 2d shows digital photographs of the blended PVA+PAA and HP-PVA/PAA membranes. The HP-PVA/PAA membrane is opaque because the spherical pores induce light scattering, whereas the PVA+PAA membrane is transparent because of the homogeneously mixed phase. The phase separation features of the HP-PVA/PAA membrane were directly identified using areal Raman spectroscopy (Fig. S11) and the relevant mapping images are shown in Fig. 2e. Strong peaks at approximately 2900 cm^{-1}, which are related to the hydroxyl (-OH) group in PVA, were observed in all the samples. The presence of a carbonyl (C=O) group in PAA is usually confirmed by peaks at approximately 1690 cm^{-1}, and no such signal was detected in pristine PVA. For the PVA+PAA-blended sample, the signals of the homogeneously dispersed molecular species, such as C=O and -OH, were present on the overall membrane surface. In contrast, the PAA-related C=O peaks for the HP-PVA/PAA membrane were predominantly observed in the internal area of the spherical pores. We further conducted atomic force microscopy (AFM) analyses to detect the presence of internal pores in HP-PVA/PAA. Clear brightness differences and height contrasts are observed in the cross-sectional AFM image of HP-PVA/PAA (Fig. S12). In contrast, the pristine PVA and PVA+PAA membranes display a lower height contrast, implying a flatter surface than that of the HP-PVA/PAA membrane. This observation strongly supports the fact that the computational study and proposed phase separation theory suitably interpreted the unique microstructural formation of HP-PVA/PAA, in which PAA-rich spherical pores were dispersed in the PVA based polymer matrix.

The FT-IR spectra of the pristine PVA, PVA+PAA, and HP-PVA/PAA membranes are shown in Fig. 3a. A broad band at 3200–3550 cm^{-1} and a peak at approximately 2850 cm^{-1}, which correspond to the O—H stretching vibration from the hydroxyl group (-OH) and C—H stretching vibration in PVA, respectively, appeared in all membrane samples [38]. When PAA was added, a distinctive peak at 1701 cm^{-1} was detected, which was attributed to the stretching band of the carbonyl group (C=O) [39]. The absorbance of the two carboxylate (COO^-) peaks was observed at 1404 and 1556 cm^{-1} (symmetric and antisymmetric stretches of COO^-, respectively), while the C=O peak in the HP-PVA/PAA membrane weakened [40]. This is because the neutralizer induced ionization of the carboxyl group to the COO^- state (as described in the inset) when the HP-PVA/PAA membrane was fabricated.

To explore the crystal structures of the HP-PVA/PAA membrane and the relevant components, X-ray diffraction (XRD) patterns (Fig. 3b) were obtained for the pristine PVA, PVA+PAA, and HP-PVA/PAA membranes. It is known that PVA is a partially crystalline polymer in which almost linear molecular chains are regularly arranged by forming self-bound hydrogen bonds between the hydroxyl groups [41,42]. Three distinctive peaks were found in the XRD patterns of all the samples, where one strong peak at 19.5° was attributed to the (101) reflection, a plane that contains the extended planar zigzag-chain direction of the crystallites, [43,44] and the other peaks at approximately 23 and 41° correspond to the (200) and (111) reflections, respectively [45,46]. The intensity of the crystalline peaks in the PVA+PAA membrane decreased [45]. This is due to the plasticizing effect of the added PAA, which improves the structural flexibility by reducing the attraction force and by increasing the distance between PVA chains [44,45]. The HP-PVA/PAA membrane exhibited weaker crystallinity than the PVA+PAA membrane, which is attributed to the nature of the semi-IPN formation, in which AA as a monomer is homogeneously dispersed between the PVA chains to be polymerized in the form of an interpenetrating crosslink between PVA and PAA, greatly reducing the crystallinity of the PVA chain [47]. The evidence of the formation of crosslinking between PVA and PAA in HP-PVA/PAA was confirmed by the increased thermal decomposition temperature in the thermogravimetric analyses (TGA), as

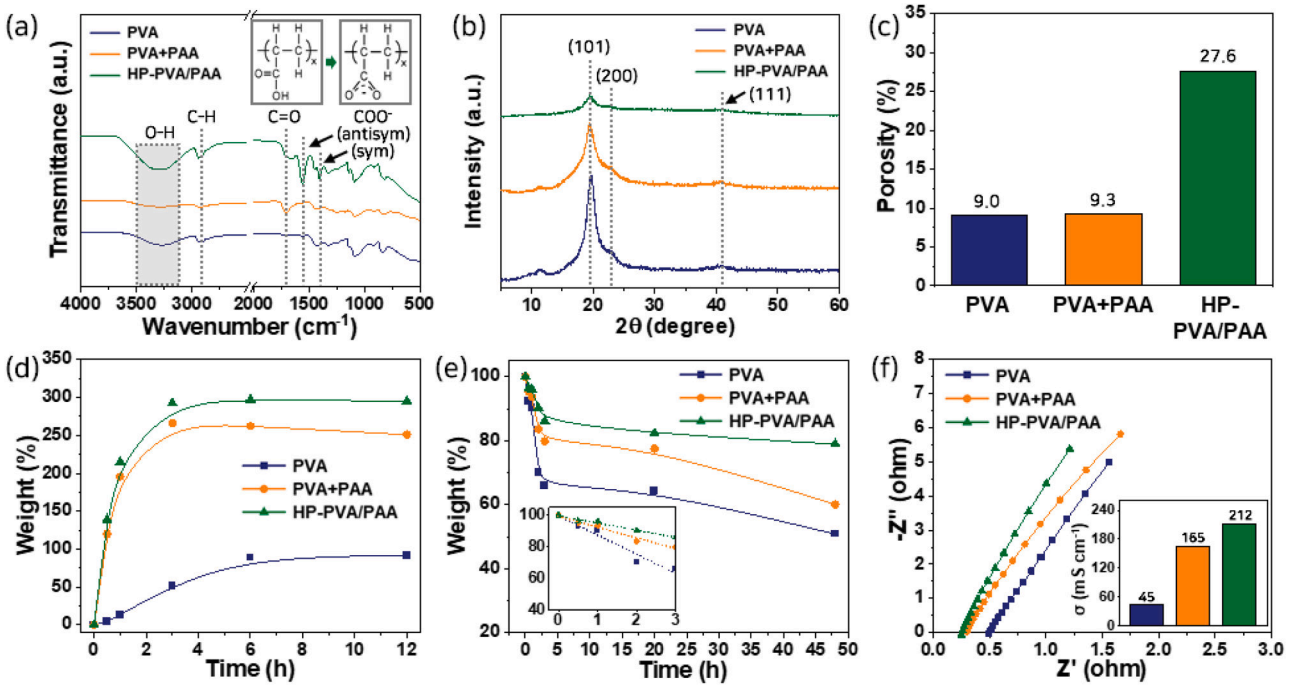


Fig. 3. Physicochemical properties of the pristine PVA, PVA+PAA, and HP-PVA/PAA membranes. (a) Fourier transform infrared spectroscopy (FT-IR) spectra, (b) X-ray diffractometer (XRD) patterns, (c) porosity values obtained using mercury intrusion porosimetry, (d) water uptake behaviors, (e) electrolyte retention capabilities, and (f) Nyquist plots and ionic conductivities (inset) of the samples.

shown in Fig. S13. It has also been reported that ionic conduction predominantly occurs through the less crystalline region in the polymer matrix because the segmental motion of local chains is more flexible than that of the crystalline region [44,45]. From these results, it is expected that HP-PVA/PAA is beneficial for fast ion transport because of its amorphous-rich polymer matrix.

The porosities of the membranes were investigated using mercury intrusion porosimetry, and Fig. S14 shows the accumulated intrusion of mercury for PVA, PVA+PAA and HP-PVA/PAA membranes [48]. The calculated porosity values of the PVA, PVA+PAA, and HP-PVA/PAA membranes were approximately 9.0, 9.3, and 27.6%, respectively (Fig. 3c). Even though the HP-PVA/PAA membrane has mostly closed internal pores, which makes it difficult for mercury to penetrate the pores, it exhibited an almost 3-fold higher porosity than the dense polymer membranes (that is, PVA and PVA+PAA). As observed using SEM (Fig. S1), the surface morphology of the HP-PVA/PAA membrane was smooth, but closed pores were formed inside the membrane. Therefore, the liquid electrolyte is trapped in the closed pores of the HP-PVA/PAA membrane, which is expected to improve long-term electrolyte retention.

Because the transportation speed of OH^- ions is greater in the liquid electrolyte than in the polymer chain due to the hopping mechanism, [49] the high absorption of the electrolyte is important for improving the overall ionic conductivity of the GPE. Fig. 3d compares the water uptake behaviors of pure PVA, PVA+PAA, and HP-PVA/PAA after 12 h of soaking in deionized water (DIW) at 25 °C. Remarkably, the PVA+PAA and HP-PVA/PAA membranes exhibited a substantial increase in water uptake as compared to that of the pristine PVA membrane, implying that the addition of PAA was beneficial for water uptake. This is closely related to the high crystallinity of the crosslinked PVA chains, which restricts the sufficient absorption of the water component, as reported in previous studies [50]. As observed in the XRD results, the decrease in crystallinity in PAA-containing membranes can impart more structural flexibility to hold water between the polymer chains. The additional ionic osmotic pressure induced by the C=O and COO^- groups in PAA can also increase the water absorbency of the GPEs [45,51]. The HP-PVA/PAA membrane exhibited approximately 18% higher water uptake after 12 h than the PVA+PAA membrane, which can be explained by the internal pore volume that accommodates additional water content.

Given the special half-open cell configuration, the water retention capability of the GPE significantly dominated the working life of the ZABs because the loss of water deteriorated the ionic conductivity, leading to early cell failure. Therefore, the water retention capability of the prepared membranes was determined by measuring the weight changes after full absorption of 6 M KOH (Fig. 3e). After 48 h of exposure of the soaked membranes to ambient conditions, PVA and PVA+PAA only maintained 50.8 and 60.1% of the GPE weight, respectively. In contrast, HP-PVA/PAA maintained in excess of 79.0% of the GPE. The improvement in electrolyte retention is consistent with the improvement in the aforementioned water uptake capabilities. The substantially restrained electrolyte loss was closely related to the microstructure of the polymer matrix. First, owing to the highly hydrophilic surface of the PAA chains, volatilization of the KOH solution was largely restricted [7,52]. Additionally, the unique closed pore structure in the polymer matrix hinders the free diffusion of water molecules and extends the diffusion distance required for water release from the polymer matrix.

Ionic conductivity is a key parameter that determines the performance of GPE based ZABs. All membrane samples were soaked in a 6 M KOH aqueous solution for 24 h before conductivity measurements were conducted. The AC impedance spectra of the PVA, PVA+PAA, and HP-PVA/PAA based GPEs are shown in Fig. 3f. The intrinsic or bulk resistance (R_b) of the GPEs was obtained by obtaining the intercept of the plot on the x-axis (Z_{re}). The calculated ionic conductivities of the GPEs are presented in the inset of Fig. 3f. PVA exhibited the lowest ionic conductivity of only 45 mS cm^{-1} owing to its low hydroxide content and water absorbency. The ionic conductivities of PVA+PAA and HP-PVA/PAA were 165 and 212 mS cm^{-1}, respectively. To the best of our knowledge, the conductivity of 212 mS cm^{-1} is one of the highest among the recently reported PVA based GPEs (Table S2), and the outstanding ionic conductivity of the HP-PVA/PAA GPE results from the synergistic effect of the following factors. First, the stronger water uptake capabilities caused by the interpenetrating PAA contribute to an increased amount of hydroxide ions (OH^-) within the PVA matrix, providing more charge carriers for facile transport [7,31,45,53]. Second, the lower degree of crystallinity of HP-PVA/PAA leads to a reduced energy barrier for ionic diffusion, thereby facilitating OH^- transport through the GPE [7,27,45]. Interestingly, we also compared GPEs comprising pores of different sizes. The ionic conductivity tends to be higher for the GPEs that comprise larger pore sizes (Fig. S15). These results indicate that the unique HP structure inside the membrane and polymer matrix of HP-PVA/PAA can accelerate ion transport through the pore-filled liquid electrolyte component, thereby increasing the overall ionic conductivity [22].

We also measured the ionic conductivities of the GPEs. The GPEs were immersed in KOH solutions of varying concentrations (0, 3, 6, 9, and 12 M) for 24 h to determine their alkaline-tolerance (Fig. S16). The ionic conductivity is determined by the number of anions inside the polymer host. The addition of KOH normally increases the amorphicity of the GPEs, resulting in improved ionic conductivity. However, the addition of excess anions reduces the ionic conductivity owing to crystallization from salt saturation, leading to the aggregation of free ions in the GPE [54,55]. This trend was also observed in PVA, PVA+PAA, and HP-PVA/PAA, but HP-PVA/PAA exhibited the highest ionic conductivity among the samples, demonstrating exceptional ionic conductivity values even when immersed in 9 and 12 M KOH (181.1 and 138.9 mS cm^{-1}, respectively). These results suggested that HP-PVA/PAA possessed superior alkaline-tolerance, which is crucial for ensuring the stability of the ZABs.

Pouch-type ZAB cells were prepared to assess the electrochemical performance of the GPEs. As shown in Fig. 4a, the ZABs were assembled with a sandwiched structure of Zn metal anode, prepared GPEs, catalyst ($Pt/C + IrO_2$)-coated CC as the cathode, and a gas diffusion layer, which was sealed with a sealed pouch where several holes were included on the cathode side of the pouch for breathing air. The HP-PVA/PAA GPE using a ZAB cell exhibited a relatively high OCV of 1.415 V (Fig. 4b). Galvanostatic voltage profiles were collected at an absolute current density of 1 mA cm^{-2} with a time span of 10 min for each charging and discharging step. The ZAB fabricated with the HP-PVA/PAA-based GPE, which has excellent electrolyte retention and high ionic conductivity, demonstrated superior rechargeability with high cycling stability for 189 cycles (Fig. 4c). In contrast, the PVA used ZAB exhibited a sudden decrease in voltage after only 44 cycles, which was significantly shorter than that of the PVA+PAA based ZAB (105 cycles). The rate performance of the as-assembled pouch-type ZABs at various current densities is shown in Fig. 4d. The HP-PVA/PAA-based GPE clearly enabled higher discharge voltages for all current densities than other GPEs, and even a discharge plateau of 0.775 V at 10 mA cm^{-2}, which is largely due to the higher ionic conductivities of the HP-PVA/PAA GPEs. Fig. 4e shows that the galvanostatic discharge performances of ZABs based on PVA, PVA+PAA, and HP-PVA/PAA GPEs were measured at 2 mA cm^{-2} under ambient conditions. All ZABs exhibited similar discharge plateaus at 1.26 V, but the HP-PVA/PAA GPE based cell exhibited the highest discharge capacity, which continued for 16.5 h. The pristine PVA and PVA+PAA-based ZABs continued their discharges for only 4.1 and 9.7 h, respectively. The specific capacities of the ZABs can be calculated from the mass of consumed Zn metal. The PVA, PVA+PAA, and HP-PVA/PAA GPE used ZABs delivered specific capacities of 168.4, 404.4, and 689.6 mAh g_{Zn}^{-1}, respectively (Fig. S17). The HP-PVA/PAA used ZABs reach peak power densities of 70.5 mW cm^{-2} (Fig. 4f), surpassing the values of the counterpart PVA and PVA+PAA used ZABs (27.7 and 46.6 mW cm^{-2}). Compared with previously reported GPE-based ZABs, the

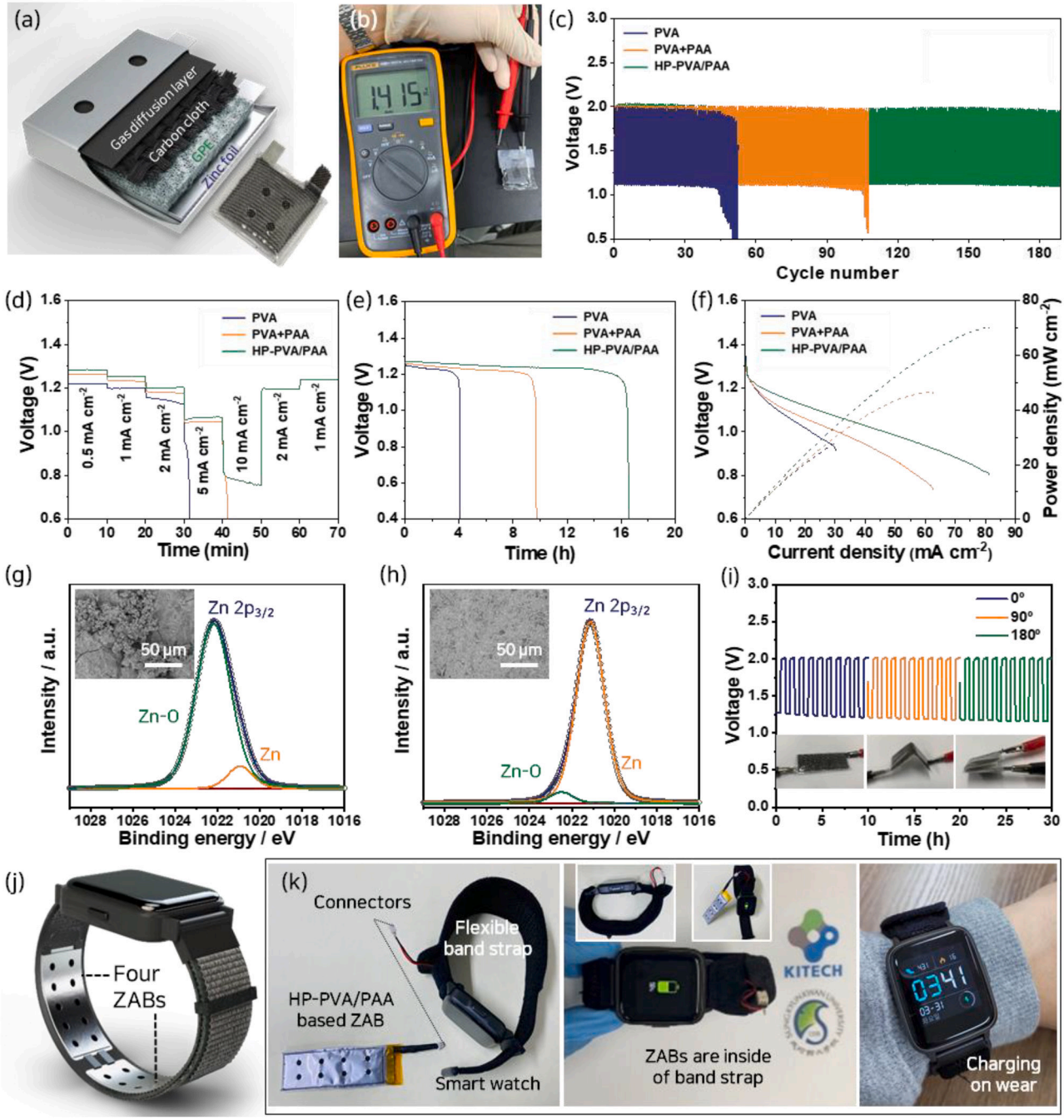


Fig. 4. (a) The schematic configuration of the pouch-type ZAB cells, (b) photographic image of the pouch-type ZAB and its open circuit voltage (OCV), (c) galvanostatic discharge and charge profiles at 1 mA cm^{-2}, (d) rate capability at different current densities, (e) specific capacity profiles, and (f) discharge profiles and corresponding power densities. High-resolution XPS Zn 2*p* spectra of the Zn anode surfaces from the used (g) PVA and (h) HP-PVA/PAA cells after 10 discharge/charge cycles (insets: *ex-situ* SEM observations of the surface of Zn anodes after 10 cycles). (i) Galvanostatic discharge and charge profiles at 2 mA cm^{-2} of the pouch-type ZAB cell at different bending angles of 0°, 90°, and 180° (j) Schematic concept and (k) various digital photographic images of the wearable device applications. A customized smart watch was powered using flexible HP-PVA/PAA ZABs inside the strap band.

developed HP-PVA/PAA GPE showed a fairly good performance to be used as a power source for wearable devices (Table S3).

The formation of dendrites on the Zn anode surface or passivation should be suppressed during battery operation to ensure the high performance of ZABs. *Ex-situ* SEM and X-ray photoelectron spectroscopy (XPS) analyses were performed after 10 discharge/charge cycles to investigate the compatibility between the GPEs and Zn anodes. The Zn 2*p* spectra of the Zn anodes that were disassembled from the used PVA and HP-PVA/PAA cells are shown in Fig. 4g and h, respectively. The deconvoluted binding energies at 1021.4 eV and 1022.3 eV are attributed to the metallic Zn and Zn-O, respectively [56]. Interestingly, the majority of the Zn 2*p* peak is associated with Zn-O binding from the PVA used cell. In contrast, the metallic Zn peak is dominant for the Zn anode of the HP-PVA/PAA used cell. Some dendrites formed on the anode of the PVA used cell, whereas they seldom formed on the Zn anode of the HP-PVA/PAA used cell (see insets of Fig. 4g and h). Dendrites are usually formed by the saturation of zincate ions on the surface of Zn, resulting in an insulating layer closely related to the formation of Zn-O bonds, as observed in the Zn 2*p* spectra. We suspect that the electrostatic interactions between the carboxylate groups along the PAA

backbone and Zn ions facilitate the formation of a homogeneous solid electrolyte interface; thereby, effectively preventing localized Zn metal nucleation and growth and thus suppressing severe Zn dendrite formation [4,57].

Considering the practical wearable applications, the high flexibility of ZABs and their reliable electrochemical performance are crucial for their use in practical wearable applications. We conducted cycling tests at various bending angles (0°, 90°, and 180°) to demonstrate the potential applications of our HP-PVA/PAA-based ZABs in wearable devices. The pouch-type ZABs incorporating HP-PVA/PAA exhibit stable cycling performances with no significant overpotential changes (Fig. 4i). The results reveal the high stability of the assembled ZAB device with robust adhesion between the GPE and the electrode layers. Additionally, the two GPE-based flexible ZABs stably operated a 3.0 V LED light (Fig. S18). Apart from just testing the pouch-type ZABs, HP-PVA/PAA-based ZABs were also demonstrated as a power source for wearable devices. A customized smart watch with four ZABs in series (2×5 cm^2) inside the strap bands was constructed (Fig. 4j), in which the smart watch (requiring 5 V for the recharge) and ZABs were connected with magnet- and wire-type connectors (Fig. S19). The prepared pouch-type ZABs, including four ZABs in series and a customized smart watch with a flexible band strap, are shown in Fig. 4k. As shown in the figures and supplemental videos, the GPE-based ZABs inserted into the band strap could act as a commercial energy source for charging the smart watch. These results demonstrate the practicability of ZABs, which show an excellent potential for an extensive variety of wearable device applications.

3. Conclusion

In summary, a HP-PVA/PAA-based GPE was successfully fabricated via nucleation and growth during the interpenetrating polymerization of AA in a PVA matrix without any inorganic or sacrificial additives. The unique lotus-like closed pores inside the HP-PVA/PAA membrane fundamentally improved the electrolyte uptake and retention as compared to the pristine or blended PVA+PAA membranes. Furthermore, the HP-PVA/PAA-based GPE showed significantly improved ionic conductivity (212 mS cm^{-1}) than the previously reported PVA-based GPEs. With these advantages, the HP-PVA/PAA GPE-based ZABs exhibited excellent cycling stability over 45 cycles, outstanding discharge rate capability, and high power output. A wearable energy storage device using these flexible ZABs as a band strap to power a smart watch was demonstrated. We believe that our novel approach can pave the way for a viable and cost-effective method for fabricating stable and flexible GPE-based energy storage systems.

4. Experimental section

4.1. Synthesis of HP-PVA/PAA and PVA+PAA-based GPEs

To synthesize the HP-PAA/PAA membrane, 10 g of PVA (Mw 110,000–148,000 g mol^{-1}, Sigma-Aldrich) was dissolved in 90 mL of DIW at 100 °C under magnetic stirring for 6 h to form a 10 wt.% PVA solution (Sol. 1). To prepare an initiator solution, 0.4 g of KPS (99.0%, Sigma-Aldrich) and 0.4 ml of hydrogen peroxide were dissolved in 10 ml of DIW (Sol. 2). Then, 2 ml of AA (99%, Sigma-Aldrich) solution was neutralized with 2.04 mL of 10 M aqueous KOH solution (neutralized to 70 mol% as compared to AA, Sol. 3), and 0.02 ml of triallylamine (crosslinking agent) was added to 40 mL of 10 wt.% PVA solution (Sol. 1) (PVA:AA = 2:1 wt.%), and gradually stirred for 30 min at 60 °C. Finally, 0.2 ml of the prepared initiator solution (Sol. 2) was added to the mixture at 100 °C overnight to complete the polymerization. The resulting polymer solution was poured into a petri dish and dried at 60 °C for 4 h. The resulting HP-PVA/PAA membrane with a thickness of ~200 μm was immersed in a 6 M KOH solution for 24 h to prepare the HP-PVA/PAA-based GPE.

To prepare the PVA+PAA-based GPE, 5.72 ml of a PAA solution (35 wt.%, Sigma-Aldrich) was added to 40 ml of 10 wt.% PVA solution (PVA: PAA = 2:1 in wt.%), and a homogeneous PVA and PAA solution was obtained after 24 h of stirring at 60 °C. The PVA+PAA solution was poured into a petri dish and then dried at 60 °C for 4 h. The obtained PVA+PAA membrane was immersed in a 6 M KOH solution for 24 h to prepare the PVA+PAA-based GPE. The PVA-based GPE was prepared using the same method but without the addition of a PAA solution.

4.2. MD simlation for demonstrating phase separation

The construction of the PVA/PAA unit cells was performed using a commercial MD software (Materials Studio, Dassault Systemes, 2020), and the simulation was performed using an open-source code (LAMMPS, Sandia Lab). PAA, PVA, water molecules, and K ions were randomly distributed in a unit cell with weight fractions of 0.11, 0.12, 0.71, and 0.06, respectively. Table S1 presents the compositional details of the PVA/PAA microstructures. To describe the degree of polymerization, four different lengths of PAA molecules, ranging from 1 to 70 repeating units, were considered. To act as a high molecular weight, the PVA chain was made with 230 monomers that were indefinitely linked beyond the periodic limit. All the acid groups in PAA were assumed to be fully ionized, and the same number of K ions (K^+) as the total repeating units of PAA were inserted for neutralization. The PVA/PAA unit cells were constructed with a density of 0.2 g cc^{-1} and contracted continuously to a density of 1.0 g cc^{-1}. Geometry optimization based on the conjugated gradient method (cutoff value of 10 kcal mol^{-1}) was applied for every 0.2 g cc^{-1} increment in density. Then, the NVT ensemble at 100 °C and the NPT ensemble at 100 °C and 1 atm were applied for 1 ns. To eliminate artifacts caused by the initial complex formation of the polymer, annealing for 3 ns at 327 °C and 1 atm and re-equilibration for 2 ns at 100 °C and 1 atm were performed under the NPT ensemble. After the density converged, the PVA/PAA unit cells in the NVT ensemble were further relaxed for 20 ns at 100 °C. A polymer consistent force field (PCFF) was adopted to model the polymeric components and K^+ and the TIP3P model was used for water molecules [58,59]. The non-covalent interactions between the water molecules and other components were described using Lennard-Jones (9–6) and the Coulomb potential was implemented in the PCFF. All the MD simulations were performed using the Verlet algorithm with a time step of 1 fs.

The concentration distributions of the K^+/PAA and PVA molecules were obtained for each equilibrated unit cell model. The unit cell was divided into 100 equal parts along each axis, and the number of atoms located in each cuboid region was counted. The values in each local region were then normalized to the total number of atoms in the polymer chains.

Additionally, the K^+, PVA, and PAA chains in the unit cell were considered molecular groups. The interaction energy between molecular components (E_{int}) can be obtained using Eq. (1):

$$E_{int} = E_{ij} - (E_i + E_j) \quad (1)$$

where E_{ij} indicates the total potential energy where two molecular groups, i and j, coexist. E_i and E_j indicate the individual potential energies of each molecular group.

4.3. Material characterization

The microstructures of the prepared samples were observed using field-emission SEM (FE-SEM; SU8010, Hitachi Co., Japan) and AFM (Park Systems XE-70, at 0.7 Hz of scan rate). The crystalline structures of the samples were analyzed using powder XRD (SmartLab, Rigaku, Japan). FT-IR was performed using a NEXUS spectrometer (Thermo Nicolet Corp., USA). The prepared samples were characterized using Raman spectroscopy (XperRam 200, Nanobase, Korea). Raman measurements were performed using a confocal micro-Raman instrument

with a backscattering configuration at room temperature. XPS measurements were performed with Kα X-ray photoelectron spectrometer (Thermofisher Scientific, U.K.) with a monochromated Al Kα X-ray source (1486.6 eV), equipped with multichannel detector. The porosities of the prepared samples were measured using a mercury porosimeter (AutoPore IV, Micromeritics Instrument Co., USA). TGA data was collected on a TGA Q500 (TA Instruments, USA), which was heated from 25 to 500 °C at a heating rate of 10 °C min^{-1} in an N_2 atmosphere.

The water uptake and retention capacities were investigated in air at 25 °C and 45% of relative humidity. The dried GPE membrane (mass w_0) was soaked in DIW for a different measurement time. Then, the DIW on the sample surface was wiped off with filter paper, and the sample was weighed again (mass w_1). To evaluate the electrolyte retention capacity, an electrolyte-saturated GPE (mass w_2) was subsequently exposed to air to accelerate water evaporation. The weight (mass w_3) of each sample was monitored over time. The water uptake (W_{up}) and electrolyte retention (W_{re}) of the GPEs were calculated using the following equations:

$$W_{up} = \frac{w_1 - w_0}{w_0} * 100\% \quad (2)$$

$$W_{re} = \frac{w_3}{w_2} * 100\% \quad (3)$$

The ionic conductivity of the GPE was measured using AC impedance spectroscopy (Zive SM6, Won-Atech, Korea). The electrolyte-saturated GPE was sandwiched between two symmetrical blocking stainless steel electrodes. The ionic conductivity (σ) was determined using Eq. (4):

$$\sigma = \frac{L}{A * R_b} \quad (4)$$

where L, A, and R_b are the thickness, geometric area, and bulk resistance of the GPE, respectively. R_b was determined from the intersection of the AC impedance plot and the real axis.

4.4. Fabrication and performance evaluation of a pouch-type ZAB

Air cathodes were prepared by spraying Pt/C + IrO_2 catalyst ink on CC and drying it for 6 h at 60 °C. The typical mass loading of the catalysts was 0.4 mg cm^{-2}. The catalyst ink was prepared by dispersing 15 mg of commercial IrO_2 and 15 mg of commercial Pt/C in a mixture of 2.4 mL DIW, 0.6 mL isopropyl alcohol (IPA), and 0.1 mL of Nafion solution (5 wt.%), followed by sonication for 1 h. Pure Zn foil (Alfa Aesar, 0.2 mm-thick, 99.9%, USA) was used as an anode. The pouch-type ZAB cells were fabricated using a layer-by-layer procedure. Typically, one piece of Zn foil and one piece of air cathode were attached face-to-face with the GPE (2.5 × 2.5 cm^2) between them. Subsequently, the cells were packed using side- and top-sealed Al-laminated materials.

To evaluate the performance of the pouch-type ZAB cells, galvanostatic discharge and charge measurements were performed using a multichannel battery-testing system (WBCS3000L, Won-Atech, Korea) under ambient conditions. The rate performance was determined at the current densities of 0.5, 1, 2, 5, and 10 mA cm^{-2}. An electrochemical workstation (Zive SM6, Won-Atech, Korea) was used to record the linear-sweep voltammetry curves (at a scan rate of 5 mV^{-1}).

CRediT authorship contribution statement

Seo Won Song: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Hongdeok Kim:** Methodology, Software, Data curation. **Seoyoon Shin:** Conceptualization, Methodology, Validation. **Seongjin Jang:** Investigation, Resources. **Jong Hyuk Bae:** Investigation, Resources. **Changhyun Pang:** Validation, Writing – review & editing. **Joonmyung Choi:** Methodology, Software, Validation, Writing – review & editing. **Ki Ro Yoon:** Conceptualization, Resources, Writing – review & editing, Visualization, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ensm.2023.102802.

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